AI vs Human Text Detection

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Today's Agenda

01 Introduction

02 Literature Review

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Introduction

As large language models grow more sophisticated and widely accessible, distinguishing between human-written and machine-generated text has become a critical challenge. This distinction matters greatly in domains like education, information search, research, legal and medical fields where trust and authenticity is essential.

Em-dashes and vague sentences are often associated with AI-generated text. But what truly distinguishes human-written text? This question guided our project choice.

Literature Review

PAPER 1

Ippolito et al. (2020) found that using both human reviewers and automatic detectors to assess the humanness of text is most ideal form of text detection.

PAPER 2

Mitchell et al. (2023) developed DetectGPT, a model that detects AI-generated text by analyzing the its own internal probability calculations to detect if the text was sampled from it. A weakness is here is, this model requires excessive computing power.

PAPER 3

Jakesch et al. (2023) found that humans often rely on misleading indicators when identifying AI text. For example, readers may assume that certain emotional tones indicate human authorship, even though AI can produce such content as well.

Our Dataset

01 Data Source

Our dataset is from Kaggle, and had approximately 487,000 text samples, each labeled as either human-written or AI-generated.

02 Sample Size

Using Python, we randomly sampled two subsets of data for the training set and test set, with each having 1,000 texts. Texts were labeled as “Human” for human-written text and “AI” for AI-generated text.

03 Topics Covered

The texts varied widely in topic, including themes such as politics, the environment, transportation, personal reflections, and general social issues. This exposed the classifier to different writing styles.

Approaches



Clean Text

- Removal of Quotes and Ciitations
- Eliminate human markers in AI



N-gram subwords

- Prefixes, suffixes, mispellings
- Lend to broader patterns



Logistic Regression

- How and where features appear in a sentence
- Stylistic choices

Evaluation Metrics

Model Evaluation Metrics:

Metric	Value
Accuracy	0.981
Kappa	0.962

113 AI	AI	0.9999281440607364	7.185593926362088...
114 AI	Human	0.483451847477633...	0.5165481525223667
115 AI	AI	0.9992121356267029	7.878643732971424...
116 Human	Human	0.005336785728692...	0.9946632142713071
117 AI	Human	0.385487126564847...	0.6145128734351522

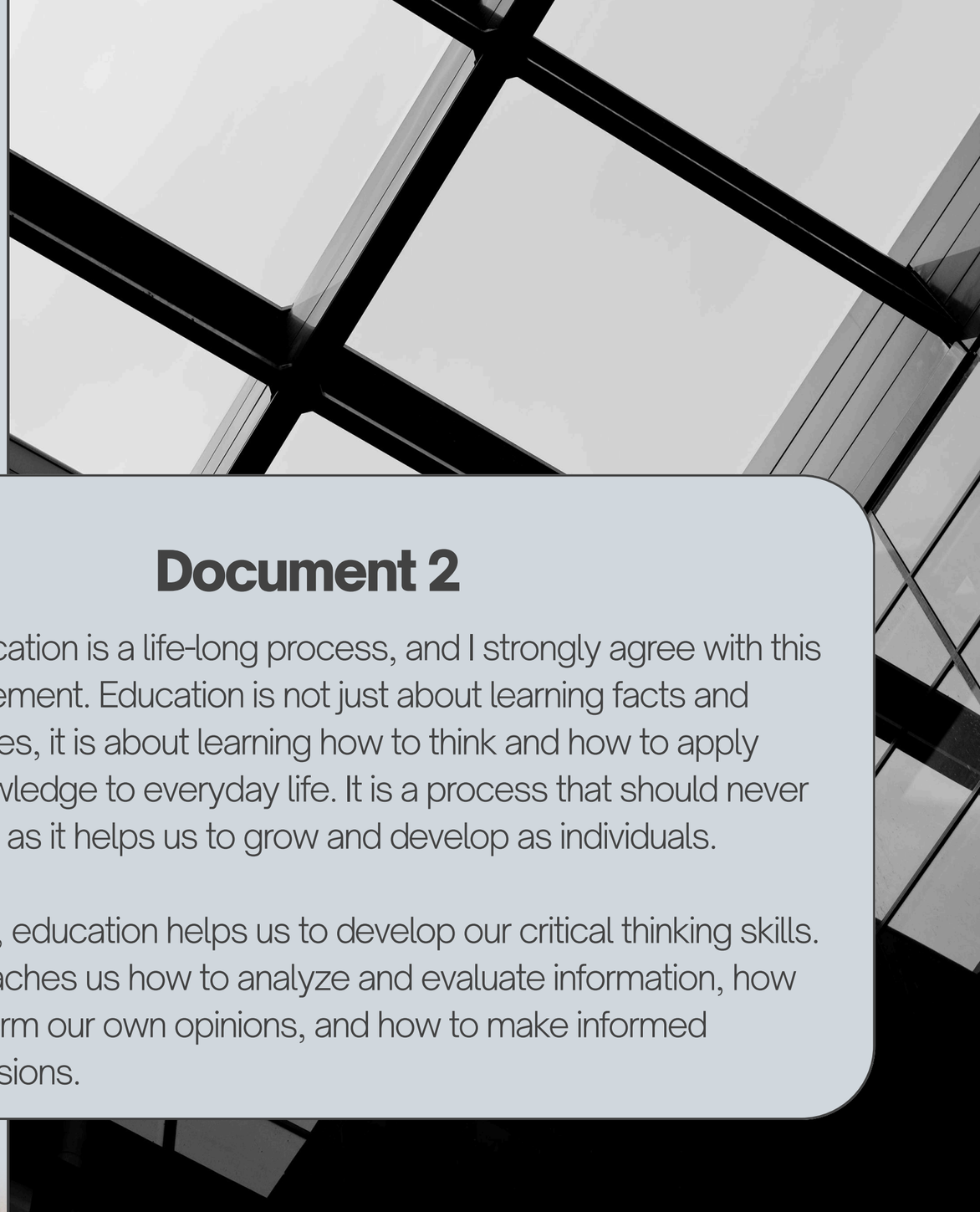
AI Features

Feature	Freque...	Fea...	Featur...
-rsb-_EOL	52	16.40...	0.3083
BOL_-lsb-	11	13.77...	0.2571
conclusion_<COMMA>	250	11.42...	0.212
<COMMA>_i	219	10.90...	0.2022
sion,__char5	261	10.28...	0.1902
con__char3	444	9.5587	0.1766
so_<COMMA>	43	9.4167	0.1739
<COMMA>_it	228	7.6151	0.1401
all__char3	361	7.4311	0.1366
who	211	7.1848	0.132
<COMMA>_like	27	6.9362	0.1274
ach__char3	170	6.8787	0.1263
ally__char4	225	6.8773	0.1263
nce__char3	209	6.6457	0.122
<COMMA>_you	54	6.6146	0.1214
<EXCLAMATIONMARK>	39	6.5633	0.1205

Human Features

Feature	Frequency	Featur...	Feature We...
ecau__char4	378	11.5148	0.2055
ecaus__char5	377	11.1733	0.1995
because	377	11.1733	0.1995
becau__char5	378	11.1733	0.1995
eca__char3	386	10.3601	0.1851
beca__char4	383	10.356	0.1851
very	156	9.9047	0.1771
n't	340	8.7364	0.1564
cause__char5	403	8.5719	0.1535
wou__char3	351	8.2621	0.148
caus__char4	409	8.2456	0.1477
cau__char3	411	7.9371	0.1422
would__char5	349	7.8505	0.1407
woul__char4	350	7.8505	0.1407
would	347	7.7796	0.1395
bec__char3	413	7.6513	0.1372

How would you classify?



Document 1

Video game violence has become a series issue in our society. Sum Bbl say it makes Bbl Mar violent, other say its just entertainment. Me? I think video games can main Bbl Mar violent.

First off, Lt look at the game themselves. Alot of the Lobular won have violent them. Bbl get hurt, killed, or blown up. its glorified. And Bbl love it. They get a rush from it. They start to think its keel to hurt Bbl. And that where it can get dangerous.

Document 2

Education is a life-long process, and I strongly agree with this statement. Education is not just about learning facts and figures, it is about learning how to think and how to apply knowledge to everyday life. It is a process that should never end, as it helps us to grow and develop as individuals.

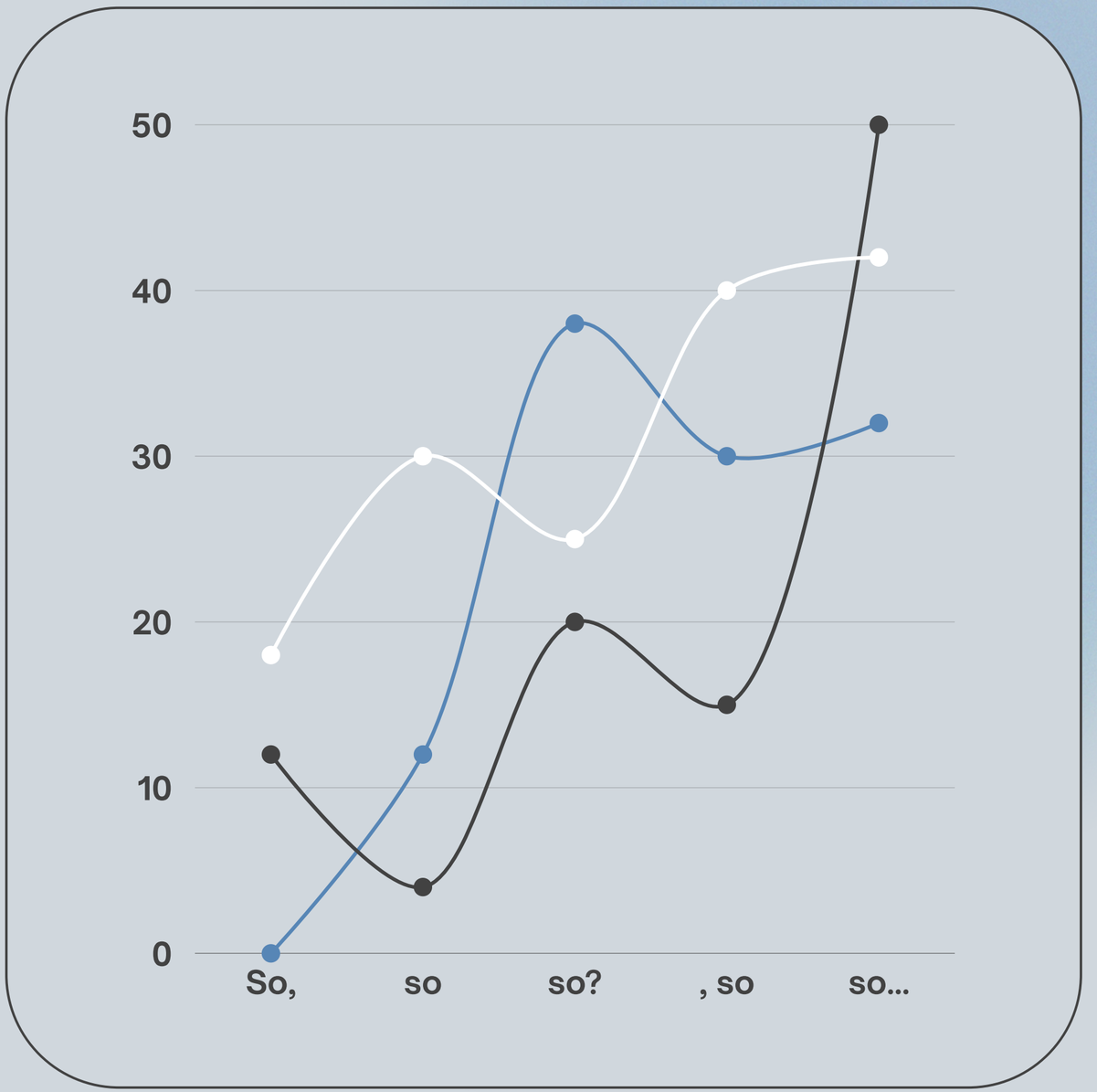
First, education helps us to develop our critical thinking skills. It teaches us how to analyze and evaluate information, how to form our own opinions, and how to make informed decisions.

“so,” vs “so”

Logistic Regression

Stylistic choices

Transition vs Conjunction



Conclusion

AI

Human

State parks provide an array of outdoor activities for people of all ages. With beaches, hiking trails, fishing, kayaking, and even camping spots, state parks offer a perfect escape from the fast paced hustle and bustle of modern day life.

First off I want to let you know that Electoral College WS a process and not a place. It was established by the founding fathers. Process consists of the selection of the electors, the meetings of the electors were they vote for president and Vice President.

The accelerating global short towards limiting car usage, Inspired by a variety or socioeconomic and environmental reasons, provides numerous advantages to society.

I believe that the face is just a natural landform because of facts. One example of why i believe it's just a natural landform is in the text it states on April 5th 1998 Mars global surveyor flew over colonia and snapped a photo ten times sharper than the original Viking photos revealing a natural landform.

Questions